

USE-CASE FLOW SPECIFICATION

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Project Title: Microlending & Peer Credit Risk Assessor

Use Case ID: UC-001

Use Case Name: Submit Loan Request

Primary Actor: Borrower

Description

This use case describes the process through which a Borrower submits a loan request through the Microlending & Peer Credit Risk Assessor. The system validates the submitted loan details, assesses the Borrower's credit risk based on their financial profile, and stores the loan request along with the calculated risk rating. The submitted request can then be reviewed by a Peer Lender.

Trigger

The Borrower selects the **Submit Loan Request** option.

Preconditions

1. The Borrower is registered in the system.
2. The Borrower is logged into their account.
3. The Borrower has created a financial profile.
4. The Borrower's financial profile contains the required information for credit risk assessment.

Postconditions

Success:

1. The loan request is successfully stored in the system.
2. The Borrower's credit risk rating is associated with the loan request.
3. The submitted loan request becomes available for the Peer Lender to view and review.

Failure:

1. The loan request is not submitted.
2. No credit risk rating is generated if the required financial information is incomplete.
3. The Borrower is informed about the missing information.

Main Success Scenario (Happy Flow)

1. The Borrower selects **Submit Loan Request**.
2. The system displays the loan request form.
3. The Borrower enters the required loan details, including the requested loan amount and repayment period.
4. The Borrower submits the completed loan request form.

5. The system validates the submitted loan details.
6. The system retrieves the Borrower's financial profile.
7. The system calculates the Borrower's credit risk rating using the available financial information.
8. The system assigns the Borrower a risk rating of **Low, Medium, or High**.
9. The system associates the calculated risk rating with the loan request.
10. The system stores the loan request in the system.
11. The system confirms that the loan request has been successfully submitted.
12. The loan request becomes available for the Peer Lender to view and review.

Alternate Flow: Incomplete Financial Profile

Condition: During the credit risk assessment, the system detects that the Borrower's financial profile is incomplete.

1. The system detects that required financial information is missing.
2. The system stops the credit risk assessment process.
3. The system does not assign a credit risk rating.
4. The system informs the Borrower that their financial profile is incomplete.
5. The system identifies the missing information that must be provided.
6. The Borrower is directed to update their financial profile.
7. The loan request is not submitted until the required financial information has been completed and a valid credit risk assessment can be performed.

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow a Borrower to create and maintain a financial profile containing income, existing debt, employment information, and repayment history.	High	Pass: The borrower can successfully save and update all required financial details. Fail: The system accepts an incomplete required profile.	A complete financial profile is required to assess the borrower's credit risk.
FR-002	Functional	The system shall compute a borrower credit risk rating (Low, Medium, or High) based on debt-to-income ratio, repayment history, and employment stability indicators.	High	Pass: A risk tier is accurately assigned using the provided indicators. Fail: The system assigns a rating when required financial data is missing.	This is the core purpose of the credit risk assessor.
FR-003	Functional	The system shall allow a Borrower to submit a loan request after a valid credit risk rating has been generated.	High	Pass: A loan request containing valid loan details is successfully submitted. Fail: A loan request is accepted without a valid risk assessment.	Peer lenders need structured loan requests with risk information before making lending decisions.
FR-004	Functional	The system shall allow a Peer Lender to view submitted loan requests, review the borrower's credit risk rating, and choose whether to fund a loan.	High	Pass: The peer lender can view available requests and successfully fund an eligible loan. Fail: The system allows funding without displaying the associated borrower risk rating.	This supports the peer-to-peer lending process and informed lending decisions.
FR-005	Functional	The system shall generate a structured repayment schedule for	High	Pass: A repayment schedule is generated after funding, and an overdue repayment	Borrowers and lenders need repayment tracking, while automated

		funded loans and trigger an automated warning when a repayment is overdue.		triggers a default warning. Fail: No schedule or warning is generated when the relevant conditions are met.	warnings help identify potential defaults early.
NFR-001	Performance & Security	Loan repayment transactions and EMI distribution shall execute within database ACID transactions to prevent double-spending and inconsistent financial records.	Medium to High	Pass: Tests confirm that transactions maintain consistency and do not result in duplicate payments under concurrent usage. Fail: Concurrent transactions cause double-spending or inconsistent balances.	Financial transactions require strong consistency and protection against duplicate or conflicting operations.
NFR-002	Reliability	The system shall ensure that borrower risk ratings, loan records, repayment schedules, and payment statuses remain available and recoverable in the event of a system failure.	High	Pass: Stored records can be recovered without loss of committed transaction data after a simulated failure. Fail: Committed financial or loan records are permanently lost or corrupted.	The platform handles financial information, so reliable storage and recovery are essential.

